Complete User Guide:
Ant Detection Application

Table of Contents

Introduction

This application helps you **automatically detect and count ants** in photographs. Think of it like teaching a computer to recognize ants by showing it examples. The app has four main sections that guide you through the complete process:

1. **Annotation** - You mark where ants are in the training images
2. **Training** - The computer learns what ants look like from your examples
3. **Testing** - You check how well the computer learned
4. **Prediction** - The computer automatically finds and counts ants in new images

Important: You must complete sections in order (Annotation → Training → Testing → Prediction).

Getting Started

Main Window Layout

When you open the app, you'll see:

- **Four tabs at the top**: Annotation, Training, Testing, and Prediction
- **Status bar at the bottom**: Shows what the app is currently doing
- **Memory usage indicator**: Shows how much computer memory is being used

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Section 1: Annotation

Purpose: This is where you annotate your image by drawing boxes around desired objects in images.

Buttons and Controls

Load Image

- **What it does**: Opens a file browser to choose images containing ants
- **When to use**: At the start of annotation - select all images you want to use for training
- **Supported formats**: JPG, JPEG, PNG
- **Tip**: Only load, annotate, and save one image at a time to avoid confusion

Zoom Controls

- **Button Options**: Zoom In, Zoom Out, and Resets Zoom
- **When to use**: Navigate a detailed image where ants are very small and difficult to annotate
- **Tip**: Use scrollbars on the side of the viewer when zoomed in to navigate image

Clear All Annotations

- **What it does**: Removes ALL boxes from the image you've annotated
- **When to use**: When you want to completely restart the annotation process
- **Warning**: This deletes everything - use carefully!

Save Annotations

- **What it does**: Saves all your annotation boxes to a file on your computer
- **When to use**:
  - After annotating each image
  - Before closing the application
  - Before switching to different image or section
- **File type**: JSON file (you'll choose where to save it)
- **Important**: Always save before moving to the next section!

Annotation Summary Box

- **What it shows**:
  - Total annotations made
  - Number of each annotation
- **Purpose**: Helps you track your progress

Delete Selected

- **What it does**: Allows you to delete the specific annotation selected in the Annotation Summary Box
- **When to use**: When you make an error while annotating and need to delete a specific annotation

Image View

- **What it does**: Allows you to view image and make annotations
- **How to use**:
  - Click and drag cursor over desired annotation area
  - Use slide bars on sides to navigate if zoomed in.

Section 2: Training

Purpose: This is where the computer learns to recognize ants based on your annotations. Think of it like a student studying from textbooks (your annotated images).

Understanding Training

The app uses a method called **Random Forest**, which creates many "decision trees" (simple rules) that vote together on whether something is an ant. More trees = more voters = potentially better accuracy (but slower).

Buttons and Controls

Load Training Images (Batch)

- **What it does**: Loads your selected images
- **When to use**: First thing when you enter the Training section
- **What you'll select**: The images you previously annotated
- **Tip**: Upload all the images you want to use to train the model at the same time

Load Training Annotations (Batch)

- **What it does**: Loads your selected annotations
- **When to use**: After you have annotated your images and uploaded them back into the Training section
- **What you'll select**: The annotations that match the images you uploaded.
- **Tip**: Upload all the annotations you want to use to train the model at the same time

View Training Pairs

- **What it does**: Opens a separate window with all of your matching images and annotations overlayed on top of each other.
- **When to use**: After you have uploaded all the images and annotations you are planning to train your model.
- **Tip**: Make sure to have consistent names for your files so the app can properly match them.
    - **Make sure to check each pair**! One incorrect match could ruin your entire training model.

Patch Size

- **What it does**: Controls the size of the patches.
    - Patches = small square windows the model slides across the image.
- **How to use**: Choose smaller or larger patches depending on ant size.
    - 16x16: Very tiny ants, high-resolution images
    - 24x24: Slightly bigger, still small
    - 32x32 (default): Normal case – most images
    - 48x48: Very large ants, low-resolution images
- **Tip**: Smaller boxes will use more memory and take longer to annotate. There is also a higher possibility the box will only see half of an ant if it is near the edge of a box
    - Make decision that is best for your situation

Detection Stride

- **What it does**: Controls the detection stride of a model
    - Stride = how far the scanning window (Patch Size) jumps each step
- **How to use**: Choose smaller or larger stride depending on situation
    - Small (4-12 px): More accurate but slower
    - Medium (16 px): Balance between speed and accuracy.
    - Large (20-30 px): Faster but less accurate and may miss ants

- **Tip**: Make selection based of preference and situation
  - Use small stride if you want maximum sensitivity because ants are small, similar to background or clumped together.
  - Use large stride if model is taking too long to run and the ants are big and obvious in the image.

Detection Scales

- **What it does**: Tells the model to scan the image at multiple zoom levels
  - Helps create a more balanced model that can detect ants of different sizes. Helpful if you upload images with different resolutions.
- **How to use**: Choose how many different zoom levels you want your model to use
  - 0.5: Zoomed out image
  - 1.0: Original image size
  - 2.0: Zoomed in
- **Tip**: Make selection based of preference and situation
  - Use more scales when the ant size varies and images come from different distances and cameras.
  - Use fewer scales when you want your model to run faster and ants are always roughly the same size.
  - Suggested to first use 0.75, 1.0, 1.25

Auto-tune Hyperparameters (recommended) **Always have this on**

- **What it does**: Hyperparameter tuning will automatically optimize random forest features based of the high, low, and middle values :
  - Number of trees: Controls how many decision trees are in your random forest model.
    - Example (default):
      - Min=25, Max=75 – Tests: [25, 50, 75]
      - Lower min= faster training but possibly less accurate
      - Higher max= potentially better accuracy but slower and uses more memory
  - Maximum depth: Controls how many levels deep each decision tree can grow (how many branches)
    - Example (default)
      - Min=6, Max=10 – Tests: [6, 8, 10]
      - Lower min= simpler trees, faster, less prone to overfitting
      - Higher max= more complex trees, can learn intricate patterns but risk overfitting
  - Min Samples Split: Minimum number of samples required to split an internal node
    - Example (default)
      - Min=5, Max=20 – Tests: [5, 12, 20]
      - Lower min= trees can split more freely, learn more details
      - Higher max= trees split only with strong evidence, more generalized
  - Min Samples Leaf: Minimum samples required at each leaf (endpoint) of the tree.
    - Example (default)
      - Min=2, Max=8 – Tests: [2, 5, 8]
      - Lower min= more detailed predictions, sensitive to individual samples
      - Higher max= smoother predictions, more robust to noise
  - Max Features: How many features to consider when looking for the best split.
    - Examples
      - Both Checked= Tests both methods, picks better one
        - Recommended default
      - Only sqrt= Uses square root of total features
        - Good for most cases, introduces randomness
      - Only log2= Uses log base 2 of total features
        - More conservative and less random
- **Tip**: Always use this feature

Train Model Button

- **What it does**: Starts the actual training process
- **When to use**: After loading data and adjusting parameters

- **What happens**:
    1. Shows progress bar
    2. Processes your annotations
    3. Trains the model
    4. Shows final accuracy metrics
- **Duration**: Depends on your parameters (seconds to 30+ minutes)
- **Tip**: Make sure to always click save model and download training report once the model has finished running.

Training Progress Area

- **What it shows**:
    - Progress bar during training
    - Status messages ("Extracting features...", "Training model...")
    - Final results when complete
- **Purpose**: Lets you know the training is working and how long to wait

Training Results Section (appears after training)

Accuracy (0-100%)

- **What it means**: Overall percentage of correct predictions
- **Good values**: Above 80%
- **Great values**: Above 90%
- **If low**: You may need more annotations or different parameters

Precision (0-100%)

- **What it means**: When the model says "that's an ant", how often is it correct?
- **High precision (90%+)**: Very few false alarms
- **Low precision (under 70%)**: Lots of false alarms (marks non-ants as ants)
- **Use case**: Important when you want to avoid counting background as ants

Recall (0-100%)

- **What it means**: Of all the real ants, how many did the model find?
- **High recall (90%+)**: Finds almost all ants
- **Low recall (under 70%)**: Misses many ants
- **Use case**: Important when you need to find every single ant

F1-Score (0-100%)

- **What it means**: Balance between Precision and Recall
- **Good values**: Above 75%
- **Use case**: Best overall metric to judge performance

Cohen's Kappa (-1 to 1)

- **What it means**: How much better the model is than random guessing
- **Above 0.8**: Excellent agreement
- **0.6-0.8**: Good agreement
- **Below 0.6**: Needs improvement

Feature Importance

- **What it shows**: Which visual features the model relies on most
- **How to read it**:
    - Taller bars = more important features
    - Shows things like "HOG gradients", "texture patterns", etc.
- **Purpose**: Understanding what the model "looks at"
- **Note**: For advanced users - you can ignore this when starting out

Save Model Button

- **What it does**: Saves the trained model to your computer
- **File type**: PKL file (pickle format)
- **When to use**:
  - After successful training
  - When metrics look good
  - Before closing the application
- **Important**: You MUST save the model to use it for predictions!

Download Training Report Button

- **What it does**: Saves the detailed training report to your computer
- **File type**: JSON file (you'll choose where to save it)
- **When to use**:
  - After successful training
  - When metrics look good
  - Before closing the application
- **Important**: Save every time you train and keep a new model, make sure to have a descriptive label

Load Model Button

- **What it does**: Opens a previously saved model
- **When to use**: Starting a new session and want to use existing model
- **Tip**: Keep multiple saved models with descriptive names

Section 3: Testing

Purpose: This section lets you see how well your trained model performs on images you have already annotated but were not used for model training. Think of it as a final exam, the model had all the study material but did not know the specific questions.

Understanding Testing

Testing shows you model:

- Precision
- Recall
- F1-Score
- mAP (mean average precision)
- True positives
- False positives
- False negatives
- Kappa score

Buttons and Controls

Load Model

- **What it does**: Loads a trained model you saved in the Training section
- **When to use**: First thing when entering the Testing section
- **File type**: PKL files
- **Required**: You must load a model before testing

Load Test Images (Batch)

- **What it does**: Loads your selected images
- **When to use**: After you have uploaded your model
- **What you'll select**: The images you previously annotated but did not show the model during training
- **Tip**: Upload all the images you want to use to test the model at the same time

Load Test Annotations (Batch)

- **What it does**: Loads your selected annotations
- **When to use**: After you have annotated your images and uploaded them back into the Testing section
- **What you'll select**: The annotations that match the images you uploaded.
- **Tip**: Upload all the annotations you want to use to train the model at the same time

View Training Pairs

- **What it does**: Opens a separate window with all of your matching images and annotations overlayed on top of each other.
- **When to use**: After you have uploaded all the images and annotations you are planning to test your model.
- **Tip**: Make sure to have consistent names for your files so the app can properly match them.
  - **Make sure to check each pair**! One incorrect match could ruin your entire test.

Detection Confidence Slider (0.0 to 1.0)

- **What it does:** Controls how confident the model needs to be before marking something as an ant.
- **When to use:** Find a balance between confidence and number of ants it counts.
  - **Lower values (0.3-0.5)**:
    - **When to use**: Want to find every possible ant, even uncertain ones
    - **Pros**: Higher recall (finds more ants)
    - **Cons**: More false positives (marks non-ants as ants)
    - **Example**: "I'd rather have some mistakes than miss any ants"
  - **Medium values (0.5-0.7)**:
    - **When to use**: Balanced approach, most common
    - **Pros**: Good balance of precision and recall
    - **Example**: Standard use case
  - **Higher values (0.7-0.9)**:
    - **When to use**: Only want very certain detections
    - **Pros**: Very high precision (few false positives)
    - **Cons**: Lower recall (misses some real ants)
    - **Example**: "I only want to count ants the model is very sure about"
  - **Recommendation**: Start at 0.8 and adjust based on your results

Distance Threshold Slider (0.005 to 0.20)

- **What it does:** Controls how close a model detection must be to an annotation to be considered a match (for evaluation only).
- **Lower values (0.005-0.020)**:
  - **When to use**: Very strict matching, need precise alignment, many ants that are clumped
  - **Pros**: Only counts near-perfect matches
  - **Cons**: Might penalize good detections that are slightly off
  - **Example**: Research requiring high precision
- **Medium values (0.025-0.10)**:
  - **When to use**: Standard evaluation
  - **Pros**: Reasonable matching criteria
  - **Example**: Most practical applications
- **Higher values (0.10-0.20)**:

    - **When to use**: Lenient matching, small annotations, few ants that are not clumped
    - **Pros**: Counts near-misses as successes
    - **Cons**: Might reward imprecise detections
    - **Example**: Very small ant images where pixel-perfect matching is hard
- **Recommendation**: Use 0.01 (1% of image size) if ants are very small and close together on the image

Run Evaluation Button

- **What it does**: Tests the model on ALL loaded images and calculates performance metrics
- **When to use**: After loading model and test data
- **What happens**:
    1. Processes all images
    2. Compares predictions to your annotations
    3. Shows detailed metrics
- **Duration**: Usually fast (seconds to a couple minutes)

Evaluation Results
The results show how well predictions match your annotations.

Precision (0-100%)
- **What it means**: Of all the detections, how many were correct?
- **Example**: 80% precision = 80 out of 100 detections are real ants, 20 are mistakes
- **High is good**: Fewer false alarms

Recall (0-100%)
- **What it means**: Of all the real ants, how many were found?
- **Example**: 90% recall = found 90 out of 100 ants, missed 10
- **High is good**: Finds more ants

F1-Score (0-100%)
- **What it means**: Balanced measure of precision and recall
- **Good values**: Above 75%
- **Use**: Best single metric to judge overall performance

Export Results Button
- **What it does**: Saves test results to a CSV file
- **When to use**: After running evaluation
- **Contents**: All metrics and per-image results
- **Purpose**: Record keeping, comparing different models

Section 4: Prediction

Purpose: This is where you use your trained model to automatically detect and count ants in NEW images (images the model hasn't seen before and you have not annotated).

This is the final goal - automated ant detection!

Buttons and Controls

Load Model

- **What it does**: Loads your trained model
- **When to use**: First thing when entering Prediction section
- **File type**: PKL files saved from Training section
- **Required**: Must have a trained model to make predictions

Load Images Button

- **What it does**: Select new images you want to analyze for ants
- **When to use**: After loading your model
- **Supported formats**: JPG, JPEG, PNG
- **Tip**: Can load multiple images at once for batch processing

Detection Threshold Slider (0.0 to 1.0)

Controls the minimum confidence score needed to mark something as an ant.

- **Lower values (0.3-0.5)**:
  - **When to use**: Don't want to miss any ants, even if unsure
  - **Pros**: Finds more potential ants (higher recall)
  - **Cons**: More false positives (mistakes)
  - **Example**: Exploratory counting where you'll manually verify results
  - **Visual result**: More green boxes on images
- **Medium values (0.5-0.7)**:
  - **When to use**: Balanced approach - most common setting
  - **Pros**: Good compromise between finding ants and avoiding false alarms
  - **Cons**: Might miss a few unclear ants
  - **Example**: Standard automated counting
  - **Visual result**: Moderate number of boxes, mostly accurate
- **Higher values (0.7-0.9)**:
  - **When to use**: Only want very confident detections
  - **Pros**: Very few false positives
  - **Cons**: Will miss some real ants (lower recall)
  - **Example**: High-stakes counting where accuracy is critical
  - **Visual result**: Fewer boxes, but very reliable
- **Recommendation**: Start at 0.8, adjust based on results

Distance Threshold Slider (0.005 to 0.20)

- **What it does:** Controls how close a model detection must be to an annotation to be considered a match (for evaluation only).
- **Lower values (0.005-0.020)**:
  - **When to use**: Very strict matching, need precise alignment, many ants that are clumped
  - **Pros**: Only counts near-perfect matches
  - **Cons**: Might penalize good detections that are slightly off
  - **Example**: Research requiring high precision

- **Medium values (0.025-0.10)**:
  - **When to use**: Standard evaluation
  - **Pros**: Reasonable matching criteria
  - **Example**: Most practical applications
- **Higher values (0.10-0.20)**:
  - **When to use**: Lenient matching, small annotations, few ants that are not clumped
  - **Pros**: Counts near-misses as successes
  - **Cons**: Might reward imprecise detections
  - **Example**: Very small ant images where pixel-perfect matching is hard
- **Recommendation**: Use 0.01 (1% of image size) if ants are very small and close together on the image

Start Detection Button

- **What it does**: Analyzes ALL loaded images and detects ants
- **When to use**: After loading model, images, and setting thresholds
- **What happens**:
  1. Shows progress bar
  2. Processes each image
  3. Draws green boxes around detected ants
  4. Counts total ants per image
  5. Shows summary statistics
- **Duration**: Depends on image count and size (seconds to minutes)

Export Summary Button

- **What it does**: Creates a CSV file with one row per image
- **When to use**: After running detection
- **Contents**:
  - Image name
  - Total ants detected
  - Average confidence score
  - Minimum confidence
  - Maximum confidence
- **Use case**: Quick overview, making charts, statistical analysis

Export Detailed CSV Button

- **What it does**: Creates a CSV file with one row per individual detection
- **When to use**: When you need complete data for each ant
- **Contents**:
  - Detection ID (unique identifier)
  - Image name
  - Detection number within image
  - Confidence score
  - Box coordinates (X1, Y1, X2, Y2)
  - Box dimensions (width, height)
  - Center point (Center_X, Center_Y)
- **Use case**: Detailed analysis, spatial statistics, tracking ant positions

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Understanding Key Terms

Machine Learning Terms

Random Forest

- **Simple definition**: A collection of many decision trees that vote together
- **Analogy**: Like asking 100 experts their opinion and going with the majority
- **In this app**: Creates multiple "trees" that each learn to recognize ants

Model

- **Simple definition**: The computer's learned knowledge about what ants look like
- **Analogy**: Like a recipe that tells the computer how to find ants
- **In this app**: Saved as a .pkl file that you can reuse

Training

- **Simple definition**: Teaching the computer using examples
- **Analogy**: Like showing flashcards to a student
- **In this app**: The computer studies your annotated images

Features

- **Simple definition**: Measurable characteristics the computer uses to identify ants
- **Examples**: Edges, textures, colors, shapes, patterns
- **In this app**: Automatically extracted using HOG, LBP, and Gabor filters

Hyperparameters

- **Simple definition**: Settings that control how the model learns
- **Examples**: Number of trees, tree depth, split criteria
- **In this app**: Controlled by the sliders in the Training section

Performance Metrics

Precision

- **Question it answers**: "When the model says it found an ant, how often is it right?"
- **Formula**: (Correct detections) ÷ (All detections)
- **High precision**: Few false alarms
- **Low precision**: Many things incorrectly marked as ants

Recall (also called Sensitivity)

- **Question it answers**: "Of all the real ants, how many did the model find?"
- **Formula**: (Correct detections) ÷ (All real ants)
- **High recall**: Finds most/all ants
- **Low recall**: Misses many ants

F1-Score

- **Question it answers**: "What's the overall balance of precision and recall?"
- **Formula**: Harmonic mean of precision and recall
- **Purpose**: Single number to judge overall performance
- **Good range**: Above 0.75 (75%)

Accuracy

- **Question it answers**: "What percentage of all predictions were correct?"
- **Formula**: (Correct predictions) ÷ (Total predictions)
- **Note**: Can be misleading if classes are imbalanced

False Positive

- **Definition**: The model detected an ant, but there isn't one
- **Also called**: False alarm, Type I error
- **Example**: Marking a piece of dirt as an ant

False Negative

- **Definition**: There's a real ant, but the model missed it
- **Also called**: Miss, Type II error
- **Example**: An ant in the image that didn't get detected

True Positive

- **Definition**: The model correctly detected a real ant
- **What you want**: Maximum true positives

True Negative

- **Definition**: The model correctly identified background (not an ant)
- **What you want**: Maximum true negatives

Computer Vision Terms

HOG (Histogram of Oriented Gradients)

- **Simple definition**: Measures the direction of edges in an image
- **What it detects**: Shapes and outlines
- **Why useful**: Ants have distinctive body shapes and leg angles
- **Analogy**: Like tracing the outline of an ant with a pencil

LBP (Local Binary Pattern)

- **Simple definition**: Measures texture patterns in small areas
- **What it detects**: Surface textures
- **Why useful**: Ant bodies have specific textures different from backgrounds
- **Analogy**: Like feeling the difference between smooth glass and rough sandpaper

Gabor Filters

- **Simple definition**: Detects patterns at specific angles and frequencies
- **What it detects**: Striped patterns, edges at different orientations
- **Why useful**: Helps detect ant legs and segmented bodies
- **Analogy**: Like looking for stripes on a zebra at different angles

Patch

- **Definition**: A small square section of an image
- **In this app**: Usually 32×32 pixels
- **Purpose**: The model examines many patches to find ants

Stride

- **Definition**: How many pixels the patch moves between examinations
- **In this app**: 16 pixels

- **Analogy**: Like moving a magnifying glass across a page

Confidence Score (or Probability)

- **Definition**: How certain the model is that something is an ant
- **Range**: 0.0 (definitely not an ant) to 1.0 (definitely an ant)
- **Example**: 0.85 means 85% confident it's an ant

Threshold

- **Definition**: The minimum confidence score to count as a detection
- **Example**: Threshold of 0.6 means ignore anything below 60% confidence

File Types

JSON (.json)

- **What it is**: Text file storing annotation data
- **Contains**: Box coordinates for all annotated ants
- **Human-readable**: Can open in text editor

PKL (.pkl)

- **What it is**: Python pickle file storing trained model
- **Contains**: Complete trained model with all learned patterns
- **Not human-readable**: Binary format for computers

CSV (.csv)

- **What it is**: Comma-separated values spreadsheet file
- **Contains**: Results data in table format
- **Opens in**: Excel, Google Sheets, or any spreadsheet program

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Tips and Best Practices

Annotation Tips

1. **Be Consistent**:
    - Use the same box size for similar-sized ants
    - Always center boxes on ants
    - Mark every visible ant, even partial ones
2. **Quality Over Quantity**:
    - 100 well-annotated images > 300 rushed annotations
    - Double-check your boxes before saving
    - Remove obviously wrong annotations
3. **Diversity in Training Data**:
    - Include images with different lighting
    - Include images with different backgrounds
    - Include images with different ant positions (walking, standing, etc.)
    - Include images with 1 ant and images with many ants
4. **Handle Edge Cases**:
    - Mark ants at image edges
    - Mark partially visible ants
    - Mark overlapping ants individually if possible

Training Tips

1. **Start Simple**:
   - Begin with 50 trees, depth 10
   - Only use auto-optimize after getting baseline results
   - Add complexity gradually if needed
2. **Interpret Your Metrics**:
   - Precision < 70%? You may have poor annotations or confusing backgrounds
   - Recall < 70%? Model might be too conservative; lower confidence threshold or add more training examples
   - Both low? Need more/better training data
3. **Training Data Requirements**:
   - Minimum: 10 images with 5-200 ants each
   - Excellent: 50+ images with varied ant counts (1000+ annotations)
4. **When to Retrain**:
   - Poor test results (F1 < 0.70)
   - New types of images/backgrounds
   - After adding more annotations
   - Significant false positives or false negatives
5. **Model Management**:
   - Save models with descriptive names: "model_v1_100trees_0.85f1.pkl"
   - Keep notes on what annotations were used
   - Keep your best-performing models

Testing Tips

1. **Always Test Before Production**:
   - Don't skip the Testing section
   - Test on the same images you trained on first
   - Test on new images second
2. **Threshold Tuning**:
   - Start at 0.8 detection threshold
   - If too many false positives: increase
   - If missing ants: decrease
   - Document what threshold works best
3. **Evaluate Per-Image Results**:
   - Look for patterns in failed images
   - Images with poor results may need more similar training examples
   - Check if certain backgrounds consistently fail
4. **Distance Threshold**:
   - Use 0.01 as default
   - Increase only if you're getting duplicate counts
   - Decrease if separate ants are being merged

Prediction Tips

1. **Pre-Process Your Images**:
   - Use consistent image quality
   - Similar lighting to training images
   - Similar backgrounds to training images
   - Consistent image resolution

2. **Batch Processing**:
    - Use folder upload for many images
    - Process during breaks (can take time)
    - Monitor memory usage for large batches
3. **Verify Results**:
    - Spot-check some images manually
    - Look at images with unusually high/low counts
    - Compare counts across similar images
4. **Export Strategy**:
    - Use Summary export for quick counts and statistics
    - Use Detailed export for spatial analysis or verification
    - Export regularly and keep organized records
5. **Handling Poor Results**:
    - Check if new images are similar to training images
    - Adjust detection threshold
    - If consistently poor, add similar images to training set and retrain

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Common Questions

Q: Why are my predictions different from my annotations? A: This is normal! The model learns patterns, not exact locations. Some variation is expected. Check if F1-score is above 0.75.

Q: What's the difference between the two CSV exports? A: Summary gives you one row per image (total counts). Detailed gives you one row per ant detected (all coordinates).

Q: Can I use this on different types of ants? A: Yes! Just train a new model with your specific ant species. The app works for any ant type.

Q: My model is too slow. What do I do? A: Reduce number of trees to 25-50. Use lower max depth (8-10). This trades some accuracy for speed.

Q: I'm getting too many false positives. Help! A: Increase detection threshold to 0.95. If that doesn't help, add more varied background examples to training.

Q: Can I train on one computer and predict on another? A: Yes! Just copy the .pkl model file to the other computer. Make sure both have the app installed.

Q: How do I know if my model is good enough? A: Check test results. F1-score above 0.75 is good, above 0.85 is excellent. Visually inspect a few predictions to ensure they look right.

Congratulations!

You now have a complete understanding of the Ant Detection Application. Remember:

- Start with quality annotations
- Train carefully with appropriate parameters
- Always test before production use
- Adjust thresholds based on your specific needs
- Export and document your results

The app is designed to make ant counting fast and accurate. With practice, you'll develop an intuition for which settings work best for your specific images and use cases.

Happy ant detecting! 🐜